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A Project Report on

"AI AUTOMATED VEHICLE INSURANCE SYSTEM"

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BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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This is to certify that the project work titled **"AI AUTOMATED VEHICLE INSURANCE SYSTEM"** is carried out by **G K MAHANIDHI (23BTRCL139), MOHAMMED AJMAL (23BTRCL005), MOHAMED ISMAIL (23BTRCL152), MOHAMED SOHAIL (23BTRCL101)** bonafide student(s) of Bachelor of Technology at the School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

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Signature of Student(s)

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ABSTRACT

Automated vehicle insurance assessment represents a critical advancement in digital finance, but it introduces complex challenges at the crossroads of computer vision, damage classification, and accurate financial estimation. This paper describes an AI Automated Vehicle Insurance System, which uses a two-stage machine learning pipeline consisting of two core modules: (1) Vehicle Damage Detection and Severity Classification using a fine-tuned MobileNetV2 deep convolutional neural network, and (2) Insurance Claim Estimation driven by an optimized ensemble regression model. The system also presents a hybrid, context-aware financial estimation scheme that dynamically bridges machine learning predictions with a rule-based fallback formula, calculating base severity, brand tier, and depreciation multipliers to ensure reliable payout outputs. Experimental findings indicate that the visual detection module successfully maps damage confidence to severity tiers, while the regression module establishes a baseline R-squared score of 0.63 with robust predictive scaling. The system is tested with a complete end-to-end implementation featuring a multi-page Streamlit frontend, Plotly visualizations, and a Python inference backend.

Keywords - vehicle insurance, damage detection, claim estimation, deep learning, MobileNetV2, ensemble regression, rule-based estimation, computer vision, Streamlit

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CHAPTER 1

1.1 INTRODUCTION

Artificial intelligence and computer vision are becoming an important part of the modern insurance industry. Applications such as automated underwriting, remote vehicle inspections, and digital claims processing are increasingly utilized to streamline operations. In the automotive insurance sector, automated assessments can make claim processing easier, faster, and more accessible for policyholders.

However, automated remote vehicle assessments introduce many technical challenges. An AI-driven insurance system must not only correctly identify whether a vehicle is damaged from an uploaded image but also accurately classify the severity of that damage. In addition, the system must translate visual evidence, vehicle attributes, and policy metadata into a reliable financial repair estimate.

Traditional claims processing relies heavily on manual human inspections and rigid estimation guidelines. These methods are inefficient, time-consuming, and subject to human bias or error. Therefore, there is a need for an accurate, adaptive, and intelligent AI-automated vehicle insurance system.

The proposed project addresses this issue by integrating deep learning and machine learning models into a single end-to-end pipeline. The system can visually classify vehicle damage severity, map contextual vehicle data, and dynamically calculate repair cost estimates using advanced regression models before presenting a complete and localized claim assessment.

1.2 MOTIVATION

The growth of digital insurance services and remote claim processing has increased the demand for faster and more user-friendly damage assessment methods. Policyholders navigating the stress of a vehicle accident, or users unfamiliar with complex insurance claim forms, often find automated, image-based reporting significantly more convenient.

At the same time, insurance providers require highly accurate estimation mechanisms to prevent inflated payouts and detect fraudulent claims. Therefore, the system must ensure both a seamless user experience and financial reliability.

1.3 AIM AND OBJECTIVE OF THE WORK

Aim:

To develop an intelligent and automated vehicle insurance assessment system that performs visual damage detection, severity classification, and context-aware claim estimation for accurate, reliable, and seamless digital insurance processing.

Objectives:

- To build an end-to-end AI-powered vehicle insurance assessment application.
- To classify uploaded vehicle images to detect the presence of physical damage.
- To categorize the visual severity of the identified damage using deep learning.
- To analyze contextual vehicle attributes, incident details and policy metadata.
- To predict accurate financial claim estimates using regression models.
- To implement a hybrid rule-based fallback mechanism for reliable cost calculation.
- To provide a user-friendly interactive interface for the claim processing.

CHAPTER 2

2.1 LITERATURE SURVEY

2.1.1 Sandler et al. proposed the **MobileNetV2** architecture for efficient image classification and computer vision tasks. The model improves traditional convolutional neural networks by introducing inverted residuals and linear bottlenecks. The MobileNetV2 model generates deep feature representations while significantly reducing computational costs and memory usage.

The model achieved state-of-the-art performance on various image benchmarks with very low latency. The main advantage of this model is that it can accurately classify high-resolution images even on constrained hardware or web platforms. This makes it highly suitable for consumer-facing insurance applications where users upload images directly from their mobile phones.

In the proposed project, MobileNetV2 is used for vehicle damage detection because it offers an optimal balance between accuracy and computational speed for real-time web deployment.

2.1.2 Wang et al. developed **CarDD**, a comprehensive framework and dataset designed specifically for vision-based car damage detection. Their research demonstrated that traditional object detection models struggle with car damage because scratches and dents have indistinct boundaries and complex textures compared to standard objects.

Unlike general-purpose vision systems, their work focuses on extracting specialized visual features to capture both global vehicle contours and fine-grained damage details. This is important in automotive insurance applications because the subtle difference between a minor scratch and a deep structural dent dictates the final repair cost.

The principles of fine-grained damage classification are used in the proposed system's computer vision module. Because of the need to differentiate damage accurately, these domain-specific visual extraction concepts are selected for this project.

2.1.3 Prokhorenkova et al. proposed the **CatBoost (Categorical Boosting) algorithm** for regression and machine learning tasks. The basic idea behind the model is a permutation-driven alternative to classic gradient boosting that prevents target leakage. The model builds

decision trees sequentially while natively processing categorical data without the need for extensive pre-processing.

In financial and insurance systems, tabular data contains heavily categorical variables. For example, attributes like vehicle manufacturer, incident type, and collision type are discrete categories, not continuous numbers. CatBoost can identify complex non-linear relationships across these categories quickly and efficiently.

The advantage of CatBoost is that it avoids the prediction shift found in older boosting methods, leading to higher R-squared scores on tabular data. Therefore, it is used in the proposed project to accurately predict the monetary insurance claim estimate.

2.1.4 Breiman proposed the use of **Random Forest**, an ensemble learning method, to improve prediction accuracy and reduce overfitting in machine learning systems. His work showed that aggregating the predictions of multiple independent decision trees often gives better and more stable results than using a single model alone.

In insurance claim severity prediction, single models are often skewed by heavy-tailed data, such as total-loss accidents which cost significantly more than average fender-benders. By combining multiple randomized trees, the ensemble system becomes more robust against extreme outliers.

The proposed project evaluates Random Forest as a foundational regression engine to map vehicle attributes to repair costs. The final claim estimation pipeline relies on these ensemble principles to establish a stable, mathematically sound financial prediction.

2.1.5 Actuarial data science studies emphasize the necessity of combining predictive machine learning with traditional deterministic models. While machine learning excels at identifying hidden patterns in historical claim data, purely data-driven models can sometimes output unrealistic financial estimates if presented with out-of-distribution inputs.

To address this, researchers suggest hybrid estimation pipelines. In the proposed project, a hybrid, context-aware estimation scheme is used. If the regression model encounters an anomaly or if certain policy variables are missing, the system dynamically shifts to a rule-based deterministic fallback.

This fallback calculates a base severity amount and applies brand tier and depreciation

multipliers, ensuring the system always outputs a financially logical estimate.

2.1.6 Existing Automated Insurance Systems

Several insurance companies such as Geico, Progressive, and Allstate have developed mobile applications for digital claims processing. These systems allow users to perform tasks such as taking photos of their damaged vehicles and uploading accident details.

However, most commercial insurance applications are not designed for instant, end-to-end automated financial estimation. They may support digital photo uploads, but they usually act only as a routing system that sends the images to a human claims adjuster for manual review. They do not combine real-time computer vision detection, damage severity classification, and dynamic ensemble regression into a single pipeline.

Because of this limitation, there is a need for a more intelligent and fully automated vehicle insurance system that can instantly verify damage, calculate repair costs, and adapt to varying vehicle constraints without human intervention.

2.2 INFERENCES DRAWN FROM LITERATURE SURVEY

- MobileNetV2 provides an optimal balance between classification accuracy and computational speed, making it highly suitable for mobile-based image uploads.
- CatBoost efficiently handles tabular insurance data with heavy categorical variables, resulting in higher regression accuracy and better R-squared scores.
- Ensemble learning methods, such as Random Forest, improve claim estimation stability by reducing the impact of extreme, high-cost outliers.
- A hybrid estimation pipeline that combines machine learning with a deterministic rule-based fallback prevents mathematically illogical financial outputs.
- Existing commercial insurance applications lack real-time, end-to-end automated financial estimation, relying primarily on human adjusters.
- An integrated pipeline combining deep learning for computer vision and ensemble regression for tabular data provides a more objective, reliable, and complete assessment than manual or single-model systems.

Sr. No.	Paper Title	Author	Year	Method Used	Contribution
1	MobileNetV2: Inverted Residuals and Linear Bottlenecks	Sandler et al.	2018	MobileNetV2	Generates deep feature representations with low latency and reduced computational cost, optimal for mobile web deployment.
2	CarDD: A New Dataset for Vision-based Car Damage Detection	Wang et al.	2022	Specialized Visual Feature Extraction	Extracts global contours and fine-grained details to accurately differentiate complex car damage textures and indistinct boundaries.
3	CatBoost: unbiased boosting with categorical features	Prokhorenkova et al.	2018	CatBoost Regressor	Natively processes discrete categorical variables without target leakage, preventing prediction shift in tabular financial data.
4	Random Forests	Breiman	2001	Random Forest (Ensemble)	Aggregates independent decision trees to improve stability and provide robustness against extreme, heavy-tailed outlier data.
5	Hybrid Machine Learning and Actuarial Models	Actuarial Studies	-	Context-Aware Hybrid Scheme	Combines predictive ML with a rule-based deterministic fallback (brand/depreciation multipliers) to ensure realistic financial outputs.
6	Existing Automated Insurance Systems	Commercial Systems	-	Manual Routing Systems	Highlights the limitation of current commercial apps, establishing the need for an instant, end-to-end automated estimation pipeline.

Table 1. Literature Survey Comparison

CHAPTER 3
SOFTWARE REQUIREMENT SPECIFICATION

3.1 INTRODUCTION

This Software Requirement Specification (SRS) document provides a detailed description of the proposed AI Automated Vehicle Insurance System. It explains the functional and non-functional requirements, system features, hardware and software requirements, and the analysis models used in the project.

The main objective of the system is to provide an accurate, automated, and intelligent platform for remote vehicle damage assessment and financial claim estimation. The system integrates computer vision-based damage detection, severity classification, context-aware regression modeling, and rule-based financial fallbacks into a single framework.

3.1.1 Project Purpose

The purpose of the project is to design and develop an intelligent and automated vehicle insurance assessment system that can:

- Analyze uploaded vehicle images to detect physical damage.
- Classify the visual severity of the identified damage.
- Process vehicle attributes, incident details, and policy metadata.
- Calculate accurate financial repair estimates using advanced regression models.
- Utilize a hybrid rule-based fallback mechanism to ensure logical financial outputs.
- Generate a complete, localized, and data-driven insurance claim assessment.

The system is designed to reduce the dependency on manual human inspections and rigid estimation guidelines, improving both the speed of claim processing and the accuracy of financial payouts.

3.1.2 Project Scope

- The system is designed to provide an automated, end-to-end multi-modal pipeline for vehicle damage assessment and financial insurance claim estimation.
- It can identify and classify specific types of vehicle damage (e.g., minor dents, surface scratches, broken glass, severe structural loss) from user-uploaded images using a computer vision perception layer.
- The system analyzes the contextual details of the incident—such as vehicle brand, manufacturing year, fuel type, and collision type—to calculate a highly accurate monetary repair estimate.
- It employs a hybrid rule-based actuarial engine as a financial guardrail to detect statistical ML anomalies and ensure all calculated payouts strictly adhere to real-world constraints like brand multipliers and age depreciation.
- The project can be used by insurance claims adjusters to expedite claim approvals, auto repair shops for quick quoting, and everyday vehicle owners seeking instant, transparent repair estimates.
- It can also be seamlessly integrated into existing auto insurance mobile applications, digital claims processing portals, or third-party risk assessment APIs..

3.2 FUNCTIONAL REQUIREMENT

Functional requirements describe the main operations and features of the system.

3.2.1 User Input & Case Initiation

- The system should allow users (adjusters, policyholders, or mechanics) to securely initiate a new claim assessment session.
- Users must provide contextual details of the vehicle and incident (e.g., Vehicle Make, Manufacturing Year, Fuel Type, Incident Type, Collision Type).
- The system should allow users to upload high-resolution images of the damaged vehicle for analysis.

3.2.2 Computer Vision Assessment (Phase 1)

- The system process the uploaded vehicle images using the fine-tuned EfficientNet deep learning model.
- The model must classify the specific type of structural or surface damage (e.g., Minor Dent, Paint Scratch, Broken Glass, Severe Structural Tear) based on visual features.
- The system should output the detected damage class and a confidence score percentage to the user

3.2.3 Contextual Feature Engineering

- The system must dynamically calculate derived features required for the regression model, such as calculating 'Vehicle Age' from the provided 'Manufacturing Year'.
- The system must handle missing or non-essential tabular inputs by imputing standard statistical medians to prevent model failure.

3.2.4 Financial Claim Estimation (Phase 2)

- The system must process the combined outputs (the visual damage classification and the tabular contextual data) through the CatBoost regression ensemble
- The model must calculate a raw, data-driven monetary repair estimate based on learned actuarial patterns

3.2.5 Hybrid Rule-Based Verification (Phase 3)

- The system must pass the raw ML prediction through a deterministic actuarial rule engine to verify financial logic.
- The engine must calculate a baseline cost based on the damage severity and apply strict multipliers (e.g., Brand Tiers, Age Depreciation, Fuel Type).
- The system must override the ML prediction if it falls outside the mathematically acceptable bounds established by the rule engine.

3.2.6 Output Generation & Visualization

- The system must present the final recommended payout to the user, including a defined low/high confidence range.

- The system must explicitly state which estimation method was used for the final quote (e.g., "Hybrid Ensemble" or "Rule-Based Fallback").
- The system should dynamically generate an interactive visual breakdown (e.g., a Waterfall Chart) detailing how the base severity cost was modified by brand, fuel, and depreciation multipliers to reach the final estimate.

3.3 NON-FUNCTIONAL REQUIREMENT

Non-functional requirements describe the quality and performance standards of the system.

3.3.1 Safety Requirements

- Vehicle images and contextual data must be processed and stored securely to protect user privacy (e.g., license plates in images).
- Financial estimates must not be generated without a complete set of valid tabular inputs and a confirmed visual damage classification.
- Proprietary actuarial logic and financial multipliers must not be exposed to unauthorized end-users.
- The system should generate audit logs for every assessment, explicitly recording whether the final quote was generated via the ML Ensemble or the Rule-Based Fallback.

3.3.2 Security Requirements

- Uploaded images must be validated for file integrity to prevent adversarial attacks on the computer vision model.
- The pre-trained ML models (efficientnet_damage_type.h5 and catboost_claim_model.cbm) must be securely stored and protected from unauthorized modification.
- Tabular inputs must be strictly validated to prevent data injection or out-of-bounds manipulation of the regression features.
- The Hybrid Rule-Based Engine must act as an immutable security layer on every transaction, hard-capping any anomalous ML predictions to prevent financial exposure.

3.3.3 Software Quality Attributes

- **Reliability**

The system should provide consistent financial estimates without frequent failures or unhandled exceptions during inference..

- **Accuracy**

The system should correctly classify the physical vehicle damage (via EfficientNet) and predict mathematically sound repair costs (via CatBoost and actuarial rules).

- **Performance**

The system should process multi-modal requests quickly, maintaining low latency during the EfficientNet image inference and CatBoost regression phases.

- **Scalability**

The architecture should support a large volume of concurrent vehicle assessments and image processing requests without severe performance degradation.

- **Usability**

The front-end interface (Streamlit) must be intuitive, allowing users to easily upload images, input data, and understand the financial Waterfall breakdown.

- **Maintainability**

The system must be modular, allowing engineers to easily retrain the CatBoost model or update actuarial multipliers as market repair costs change over time.

3.4 SYSTEM REQUIREMENT

3.4.1 Software Requirements

● Windows 10/11 or Linux distributions (Ubuntu 20.04+)

● Python 3.10 or above

● Streamlit

● Pandas, Numpy, Matplotlib, Seaborn, Plotly

● TensorFlow / Keras

● CatBoost, Scikit-learn

● Visual Studio Code

3.4.2 Hardware Requirements

● 8 GB RAM (Minimum) / 16 GB RAM (Recommended)

● Intel Core i5 / AMD Ryzen 5 or higher

● NVIDIA GeForce GTX 1650 or above (CUDA Enabled GPUs)

● 256 GB SSD and 1 TB HDD

3.5 ANALYSIS MODELS

The project utilizes a multi-modal architecture combining deep learning computer vision, advanced ensemble regression, and deterministic actuarial mathematics for secure and automated insurance processing.

3.5.1 Visual Perception Model (EfficientNet-B0)

EfficientNet-B0 is utilized as the primary deep learning vision architecture to analyze user-uploaded vehicle images. Using compound scaling, it extracts fine-grained spatial features to

accurately classify the physical severity of the incident (e.g., paint scratch, deep dent, broken glass, or missing structural parts) while maintaining exceptional computational efficiency..

3.5.2 Claim Estimation Regression Engine (CatBoost)

CatBoost (Categorical Boosting) is employed to process the heterogeneous tabular data surrounding the incident. It correlates discrete categorical inputs (vehicle brand, collision type, fuel type) with continuous variables (vehicle age, policy deductible) to predict a base monetary repair cost without target leakage.

3.5.3 Hybrid Actuarial Verification Engine

A deterministic rule-based framework operates alongside the machine learning models to act as a financial guardrail. It applies strict, hardcoded actuarial mathematics—such as brand premium multipliers and age-based depreciation curves—to the ML predictions to ensure the final output is realistic and bounded.

3.5.4 Contextual Feature Engineering Pipeline

An algorithmic data-preparation pipeline is used dynamically at runtime to synthesize raw user inputs into advanced predictive signals (e.g., converting "manufacturing year" into "vehicle age", or calculating "premium-to-deductible ratios") before passing the tensor to the CatBoost model.

CHAPTER 4

DESIGN AND MODELLING OF SYSTEM

4.1 Overall System Design

The proposed system is designed using a multi-layer V.I.E.W. (Visual Inference and Ensemble Weighting) architecture that combines deep learning computer vision, tabular machine learning regression, and a deterministic actuarial rule engine.

The system operates in three distinct layers. It begins with the user providing incident context and vehicle imagery. The **Perception Layer** extracts visual damage features, the **Intelligence Layer** evaluates the financial regression variables, and the **Action Layer** applies strict actuarial bounds before returning a final, context-aware monetary estimate to the user.

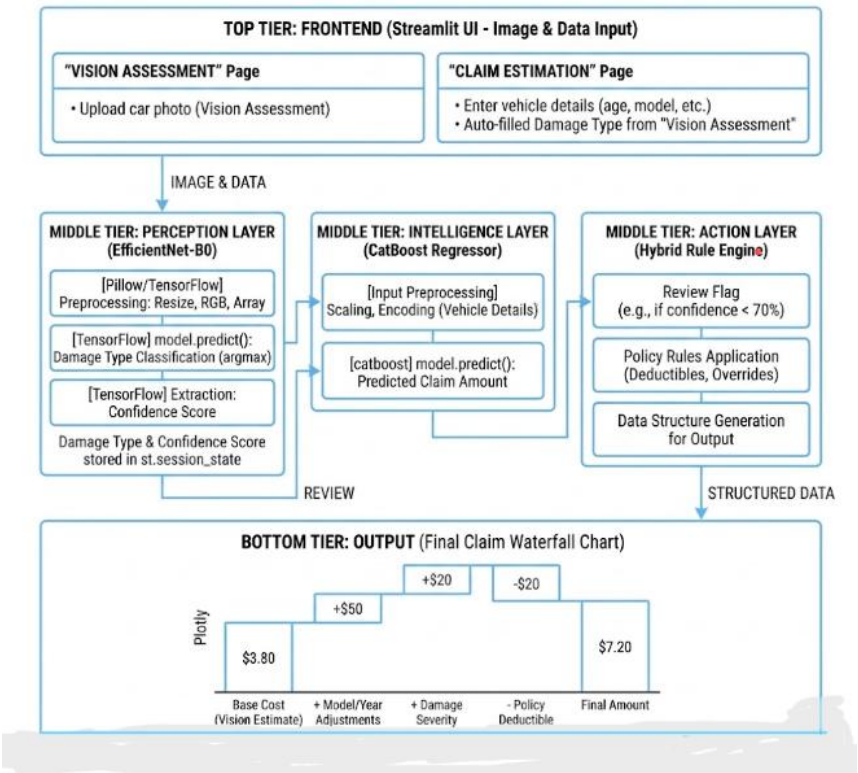

Fig1. System Architecture

This figure shows the complete architecture of the proposed system as 3 tiers and including frontend.

4.2 Multi-Modal Assessment Pipeline Flow

The assessment flow begins when the user uploads a photograph of the damaged vehicle and inputs contextual tabular data (e.g., manufacturing year, brand, fuel type).

The pipeline splits this data into two parallel streams. The image is routed to the visual perception module to classify the physical severity (e.g., Minor Dent, Broken Glass). Simultaneously, the tabular data is routed to a dynamic feature engineering script. Once the vision model outputs the damage classification, it is appended to the tabular data as a new feature. This enriched dataset is then passed into the regression ensemble to predict the raw claim cost, which is finally verified by the hybrid actuarial engine.

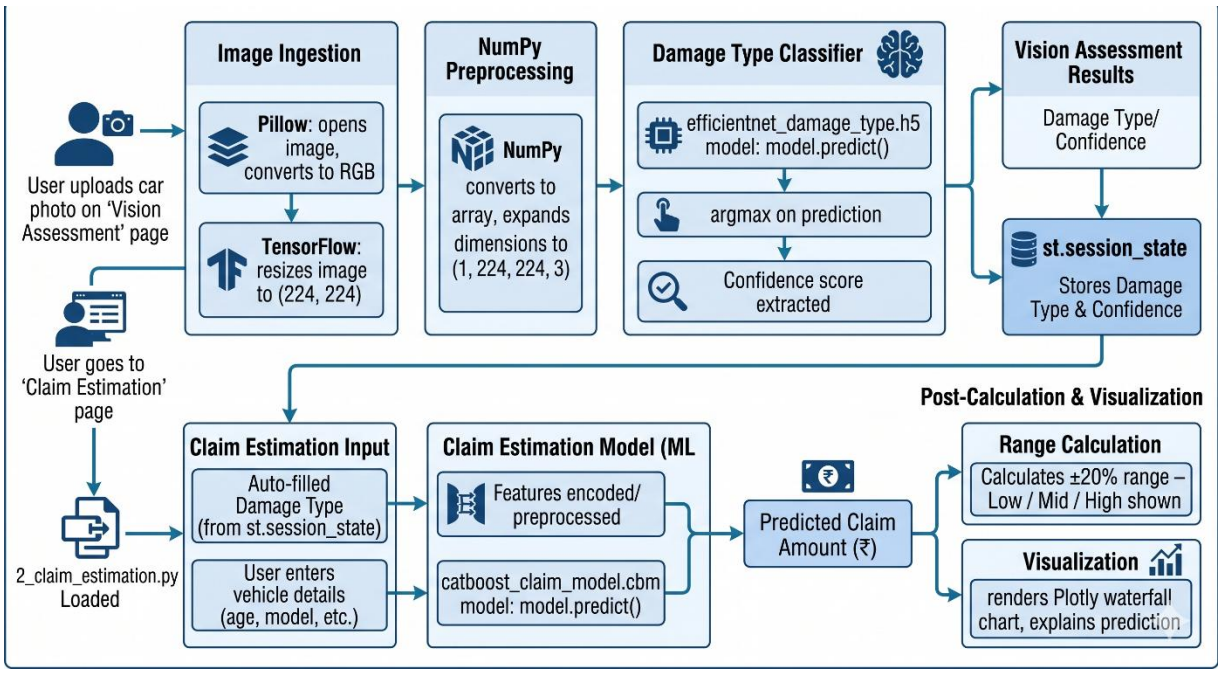

Fig2: Pipeline flow

This figure shows the flow from image input to damage classification and Claim Estimation processing.

4.3 Visual Perception Model (Phase 1)

The visual perception module relies on the EfficientNet-B0 convolutional neural network. Unlike traditional object detection models, this architecture utilizes compound scaling to balance network depth, width, and resolution.

The system reads the pixel matrix of the uploaded image and processes it through inverted bottleneck layers to identify complex textural deformations—such as the indistinct boundaries of a dent or the jagged edges of a structural tear. The model outputs the predicted damage class alongside a percentage-based confidence score.

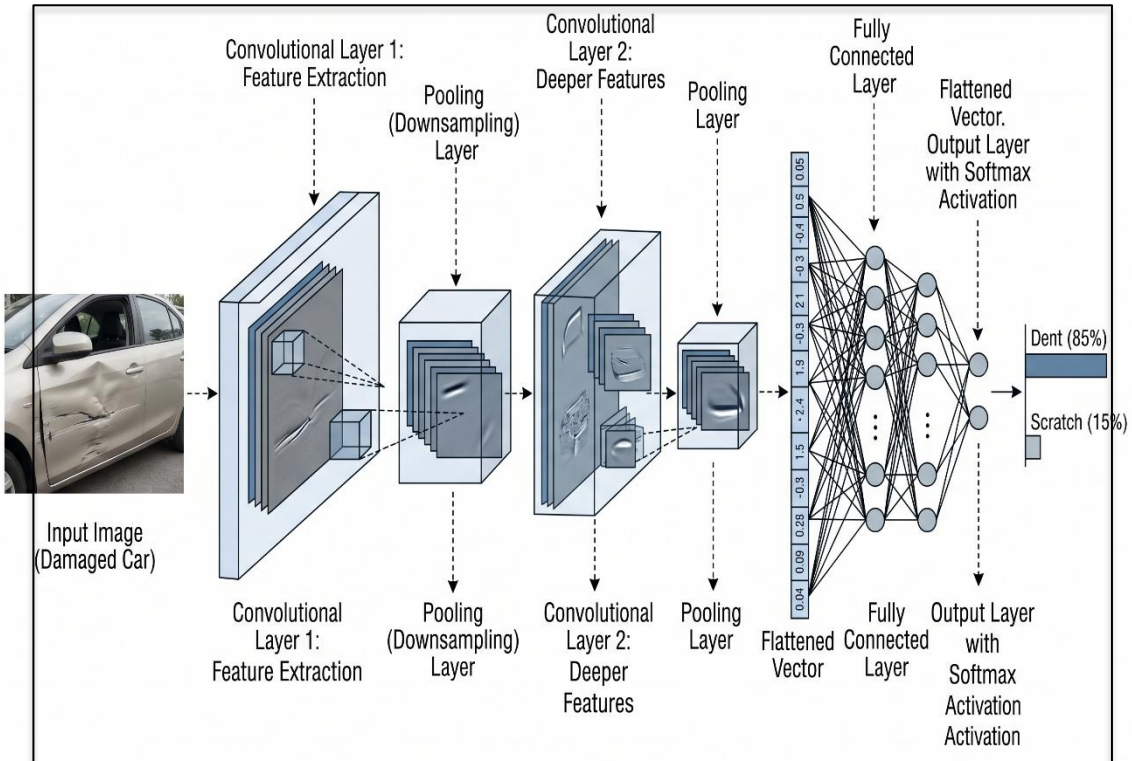

Fig3: Visual Perception Model

This figure shows the similarity score distribution between genuine speakers and imposters.

4.4 Claim Estimation Regression Engine (Phase 2)

To handle the heterogeneous data typical of insurance claims, the system utilizes CatBoost (Categorical Boosting).

The model maps the complex, non-linear relationships between the driver's profile, the vehicle mechanics, and the visually detected damage severity. Because insurance data contains heavy categorical variables (e.g., Car Make, Incident Type), CatBoost uses ordered boosting and

oblivious decision trees to prevent target leakage and prediction shift, resulting in a highly accurate baseline financial estimate.

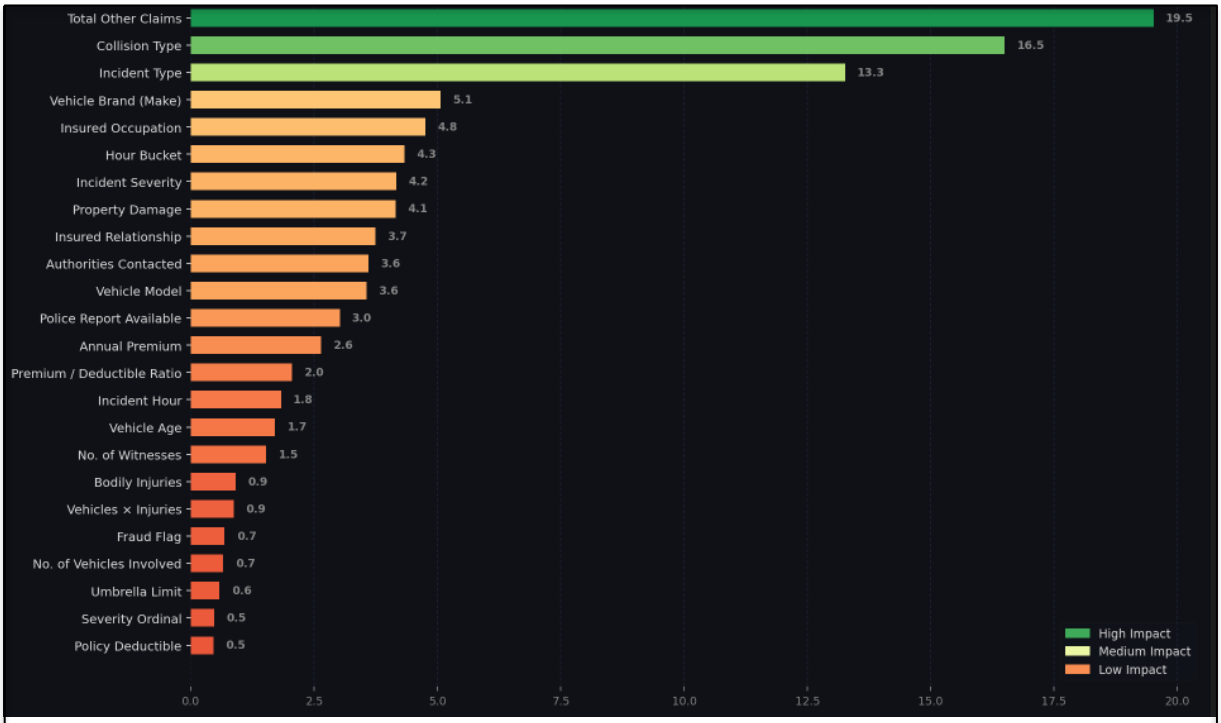

Fig4: Feature Importance

This figure shows how different features affect the claim estimate amount.

4.5 Hybrid Actuarial Verification (Phase 3)

To prevent the machine learning models from generating mathematically anomalous outputs (such as predicting a repair cost higher than the vehicle's total value), a deterministic verification step is enforced.

The raw output from the CatBoost model is passed into a rule-based engine. This engine applies hardcoded multipliers based on the vehicle's brand tier (e.g., luxury vs. economy) and age depreciation. If the ML prediction violates the actuarial bounds, the system triggers a mathematical fallback to ensure the final output remains financially logical.

Fig5: Actuarial Breakdown

This figure shows the waterfall chart generated to represent the estimate amount.

CHAPTER 5

ALGORITHM AND METHODS

5.1 Data Acquisition

The V.I.E.W. system utilizes a multi-modal data ingestion strategy, collecting both unstructured visual data and structured tabular data during the assessment stage.

Visual Perception Data: The foundational image dataset for training the vision model is based on the VehiDE dataset, comprising annotated images of real-world vehicle collisions. During active inference (via the Streamlit frontend), the system accepts user-uploaded images (JPEG/PNG format) representing the damaged vehicle.

Tabular Actuarial Data: The foundational regression model was trained using a comprehensive Kaggle Vehicle Insurance Claim dataset. During inference, the system collects real-time context from the user through the UI. The collected transaction-related features include:

- Vehicle Brand and Manufacturing Year
- Incident Type (e.g., Single vs. Multi-vehicle Collision)
- Collision Type (Front, Rear, Side)
- Policy Deductible and Annual Premium
- Hour of the Incident
- Contacted Authorities (Police, Ambulance, Fire)

5.2 Data Pre-processing

Before passing data to the machine learning models, both the images and the tabular inputs undergo rigorous cleaning and transformation.

Image Pre-processing: To meet the architectural constraints of the EfficientNet-B0 network, all input images are resized to a fixed matrix of 224×224 pixels. The pixel values are normalized to a standard scale to ensure stable gradient descent during inference.

Tabular Feature Engineering: Raw user inputs are algorithmically transformed into higher-order predictive signals before reaching the CatBoost ensemble. Key engineered features

include:

- **Vehicle Age:** Calculated dynamically by subtracting the manufacturing year from the current temporal year.
- **Hour Buckets:** Continuous time data is grouped into distinct categorical phases (Morning, Afternoon, Evening, Night).
- **Financial Ratios:** The premium_deductable_ratio is calculated to assess the policyholder's financial risk profile.
- **Collision Energy Proxy:** Calculated by interacting the number of vehicles involved with reported bodily injuries to estimate the physical severity of the crash.
- Missing variables are safely imputed using domain-specific medians or explicitly categorized as "Unknown" to prevent target leakage.

5.3 Data Augmentation

To prevent the EfficientNet model from overfitting to the specific lighting conditions or camera angles present in the VehiDE training dataset, a robust data augmentation pipeline is applied during the training phase. The following spatial and pixel-level transformations are utilized:

- **Random Horizontal Flipping:** Ensures the model recognizes damage regardless of the car's orientation.
- **Random Rotation (15%):** Accounts for users taking photos at tilted or awkward angles.
- **Random Zoom (15%):** Simulates varying distances between the camera and the vehicle.
- **Random Contrast Enhancement (20%):** A critical augmentation that forces the neural network to rely on structural deformations rather than just lighting, making it significantly better at detecting subtle dents hidden in dark shadows.

5.4 Proposed V.I.E.W. Pipeline Network

The proposed system combines a deep learning convolutional neural network, a gradient boosting algorithm, and a deterministic mathematical engine in sequence:

- **EfficientNet-B0** for Visual Damage Classification
- **CatBoost Regressor** for Base Claim Estimation
- **CatBoost Classifier** for Tiered Risk Assessment
- **Hybrid Actuarial Engine** for Financial Bounding

The primary objective function optimized by the CatBoost Regressor during the training phase is the Mean Absolute Error (MAE), which prevents massive financial outliers from skewing the model. It is calculated using:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

(Where n is the number of claims, y_i is the actual claim amount, and $\hat{Y}_i$

is the predicted amount).

Once the machine learning prediction is localized to Indian Rupees (INR), the Hybrid Actuarial Engine applies deterministic guardrails to finalize the payout. The final hybrid score is calculated using:

$$Claim_Final = Estimate_ML \times \Delta_Brand \times (1 - \delta_Depreciation)^Age$$

(Where Δ_{Brand} represents the categorical brand premium multiplier, and $\delta_{\text{Depreciation}}$

represents the fixed annual depreciation rate applied over the vehicle's age).

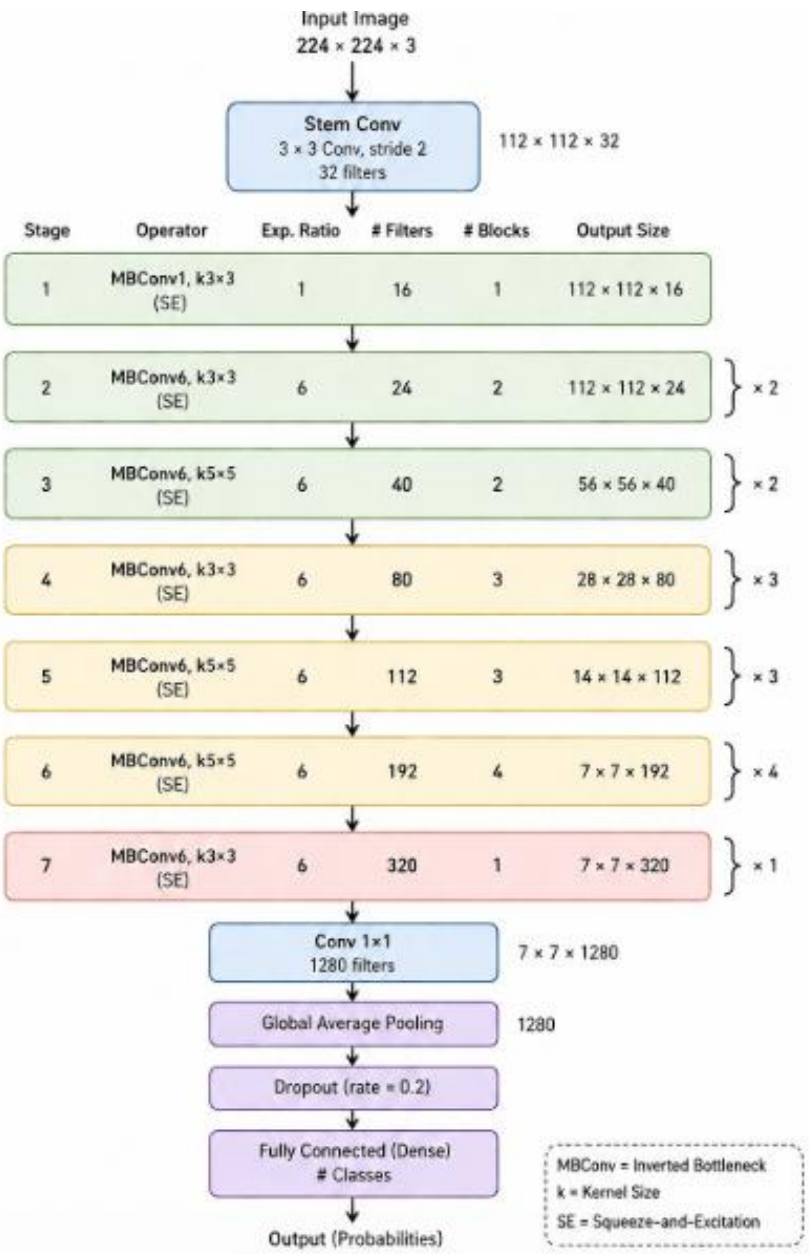

Fig6: EfficientNetB0 Architecture

This figure shows the architecture of the EfficientNetB0 model used for Damage classification.

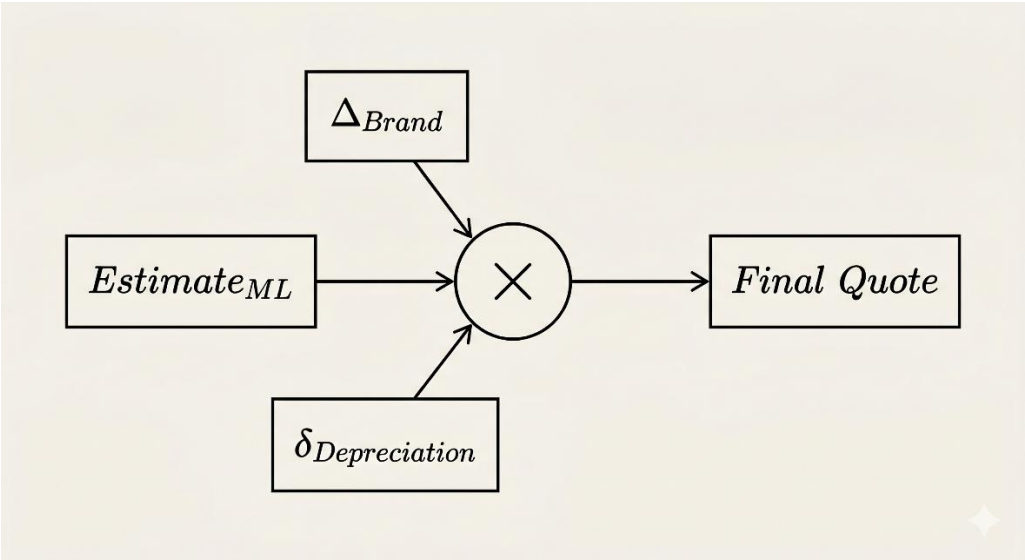

Fig7: Hybrid Engine Calculation

This figure shows how the fraud detection model combines Isolation Forest and Random Forest outputs to generate the final fraud score.

CHAPTER 6

IMPLEMENTATION & TESTING

6.1 Frontend Implementation

The frontend of the proposed V.I.E.W. system is developed using **Streamlit**, a Python-based web application framework. The user interface is designed to be highly intuitive, allowing insurance adjusters or vehicle owners to perform complex multi-modal assessments without writing code.

The frontend provides the following core pages and features:

- **System Overview Page:** A landing page detailing the V.I.E.W. architecture.
- **Phase 1: Vision Assessment:** An interface with a drag-and-drop file uploader for vehicle images.
- **Phase 2: Claim Estimation:** A dynamic form to input contextual variables (e.g., Vehicle Brand, Manufacturing Year, Incident Type, Policy Deductible).
- **Interactive Dashboard:** A Plotly-powered interface that renders the final estimation results.

The frontend dynamically handles user inputs and routes them to the underlying ML models. It displays crucial assessment details, including:

- Detected physical damage class (translated to English insurance terms)
- Vision model confidence score (%)
- The engine used for the final calculation (e.g., "Hybrid Rule-Based" vs "ML Ensemble")
- The recommended financial payout in Indian Rupees (₹), including low/high ranges.
- An interactive Waterfall Chart visually breaking down base costs, brand premiums, and age depreciation.

6.2 Backend & Logic Implementation

The backend processing is handled natively within the Python ecosystem, utilizing modular utility scripts (hybrid_engine.py, visualizer.py) called by the Streamlit server. The backend is responsible for receiving the heterogeneous data, passing it through the multi-modal pipeline, and returning a bounded financial quote.

When a user initiates an assessment, the backend executes the following sequence:

1) **Image Preprocessing:** Resizes and normalizes the uploaded image for the CNN.
2) **Vision Inference:** Passes the tensor to the EfficientNet-B0 model to extract the damage class.
3) **State Management:** Saves the detected visual class into st.session_state to securely pass it to the estimation module.
4) **Feature Engineering:** Synthesizes the raw tabular inputs (e.g., converting 'Manufacturing Year' to 'Vehicle Age', calculating 'Premium/Deductible Ratio').
5) **Regression Inference:** Feeds the engineered 24-feature dataset into the CatBoost Regressor to predict the raw USD estimate, immediately converting it to INR (₹).
6) **Actuarial Bounding:** Routes the ML prediction to the Hybrid Engine to apply deterministic mathematical constraints.
7) **Visualization Generation:** Compiles the mathematical steps into a JSON object for the Plotly frontend to render.

6.3 Data persistence and State Management

Because the system is designed for instant, real-time assessment, data storage focuses heavily on rapid state management, model metadata, and claim logging.

Model & Metadata Storage: The system loads highly optimized pre-trained model states into cached memory upon startup:

- efficientnet_damage_type.h5 (Vision Weights)
- catboost_claim_model.cbm (Regression Trees)
- model_features.json (Categorical mappings and feature lists)

Session State Management: Streamlit's native session state is utilized as an in-memory database during the user's active workflow. It strictly enforces sequential logic, preventing users from accessing the Phase 2 Claim Estimation module until Phase 1 (Vision Assessment) has successfully populated the state with a valid damage classification.

Assessment Audit Logs (SQLite): For enterprise deployment, a lightweight SQLite database is utilized to persist assessment history for auditing. The Claim_Records table stores:

- Assessment ID and Timestamp
- Vehicle Make, Year, and Fuel Type
- Detected Visual Damage Class
- Raw ML Estimation (₹)
- Final Hybrid Payout (₹)
- Rule Engine Trigger Status (Boolean)

This setup ensures that all critical technical states and financial assessments are securely generated, bounded, and logged for future reference.

6.4 Model Training Process

The training process for the V.I.E.W. pipeline is divided into two distinct regimes: the Deep Learning Vision model and the Tabular Ensemble model.

Visual Perception Training (EfficientNet-B0):

The computer vision model is trained using the VehiDE dataset to classify structural vehicle damage. To prevent catastrophic forgetting and optimize feature extraction, a two-phase transfer learning approach is utilized:

- **Phase 1 (Head Training):** The pre-trained ImageNet base is frozen, and only the top classification dense layers are trained.
- **Phase 2 (Deep Fine-Tuning):** The top 40 layers of the EfficientNet base are unfrozen to allow the model to learn dataset-specific textural variations (e.g., distinguishing deep dents from surface scratches).

The training process utilizes the following hyperparameter configuration:

- **Optimizer:** Adam
- **Learning Rate:** 1×10^{-3} (Phase 1) reduced to 1×10^{-4} (Phase 2)
- **Batch Size:** 32
- **Epochs:** 8 (Phase 1) + 20 (Phase 2)
- **Loss Function:** Categorical Cross-Entropy
- **Advanced Callbacks:** ReduceLROnPlateau (to navigate local minima) and EarlyStopping (restoring best weights).

```
--- PHASE 1: Training Classification Head ---
Epoch 1/10
939/939 ━━━━━━━━━━━━━━━━━━━━ 331s 348ms/step - accuracy: 0.6036 - loss: 1.0705 - val_accuracy: 0.6655 - val_loss: 0
.9268
Epoch 2/10
939/939 ━━━━━━━━━━━━━━━━━━━━ 336s 358ms/step - accuracy: 0.6490 - loss: 0.9513 - val_accuracy: 0.6708 - val_loss: 0
.9021
Epoch 3/10
939/939 ━━━━━━━━━━━━━━━━━━━━ 349s 372ms/step - accuracy: 0.6583 - loss: 0.9296 - val_accuracy: 0.6799 - val_loss: 0
.8794
Epoch 4/10
939/939 ━━━━━━━━━━━━━━━━━━━━ 332s 354ms/step - accuracy: 0.6605 - loss: 0.9199 - val_accuracy: 0.6877 - val_loss: 0
.8695
Epoch 5/10
939/939 ━━━━━━━━━━━━━━━━━━━━ 319s 340ms/step - accuracy: 0.6610 - loss: 0.9195 - val_accuracy: 0.6878 - val_loss: 0
.8647
Epoch 6/10
347/939 ━━━━━━━━━━━━━━━━━━━━ 2:53 292ms/step - accuracy: 0.6764 - loss: 0.8940
```

Fig8: EfficientNetB0 Vision Model Training

Claim Estimation Training (CatBoost Ensemble):

The intelligence layer is trained on a comprehensive auto insurance dataset enriched with dynamic feature engineering. To achieve robust financial bounding, a dual-objective training strategy is executed:

1. **CatBoost Regressor (Monetary Quote):** Trained to minimize Mean Absolute Error (MAE) rather than RMSE. This mathematically prevents massive, heavy-tailed financial outliers from skewing standard predictions. *(Parameters: Iterations = 4000, Learning Rate = 0.012, Depth = 8, L2 Leaf Regularization = 3.0)*.
2. **CatBoost Classifier (Risk Tiering):** Trained using a MultiClass objective to categorize claims into Low, Medium, and High severity tiers based on 33rd/66th percentiles, optimizing for a balanced Macro-F1 score.

The tabular models are trained on a robust 24-feature engineered dataset comprising elements such as:

- Vehicle Manufacturer and Derived Vehicle Age
- Incident Severity and Collision Type
- Premium-to-Deductible Ratios
- Collision Energy Proxies (Interaction between bodily injuries and vehicles involved)
- Hourly temporal buckets

```
--- PHASE 2: Fine-Tuning ---
Epoch 1/15
939/939 ━━━━━━━━━━━━━━━━━━━━ 0s 321ms/step - accuracy: 0.6207 - f1_score: 0.6149 - loss: 1.0560WARNING:absl:You are
 saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is cons
idered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.sa
ving.save_model(model, 'my_model.keras')`.
939/939 ━━━━━━━━━━━━━━━━━━━━ 362s 380ms/step - accuracy: 0.6635 - f1_score: 0.6570 - loss: 0.9327 - val_accuracy: 0
.7278 - val_f1_score: 0.7193 - val_loss: 0.7709
Epoch 2/15
939/939 ━━━━━━━━━━━━━━━━━━━━ 0s 363ms/step - accuracy: 0.7180 - f1_score: 0.7129 - loss: 0.7790WARNING:absl:You are
 saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is cons
idered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.sa
ving.save_model(model, 'my_model.keras')`.
939/939 ━━━━━━━━━━━━━━━━━━━━ 403s 429ms/step - accuracy: 0.7265 - f1_score: 0.7223 - loss: 0.7554 - val_accuracy: 0
.7500 - val_f1_score: 0.7436 - val_loss: 0.7068
Epoch 3/15
939/939 ━━━━━━━━━━━━━━━━━━━━ 0s 368ms/step - accuracy: 0.7489 - f1_score: 0.7454 - loss: 0.6992WARNING:absl:You are
```

Fig9: Fine-Tuning of the Model

6.5 Testing Procedure

The V.I.E.W. pipeline is rigorously tested using a diverse holdout dataset of vehicle collision images and a wide variance of simulated tabular claim scenarios. This ensures both predictive accuracy and absolute financial safety before deployment.

The following testing edge-cases and scenarios are evaluated:

1. **Standard Claim Assessment:** Clear visual damage on a standard economy vehicle with expected tabular context.
2. **Total Loss Scenario:** High-severity structural damage requiring maximum policy payout predictions.
3. **Ambiguous Visual Input:** Images with heavy shadows, reflections, or awkward camera angles to test the robustness of the EfficientNet data augmentation.
4. **Data Anomaly/Missing Inputs:** Scenarios with incomplete tabular data (e.g., missing policy deductible or unknown vehicle age) to test the dynamic imputation pipeline.
5. **Actuarial Boundary Testing:** Extreme outlier ML predictions purposefully injected to verify that the Hybrid Rule-Based Engine correctly "clamps" the final financial quote to realistic bounds.
6. **Multiplier Logic Testing:** Cross-referencing identical visual damage on a luxury sedan versus an economy hatchback to ensure the Δ_{Brand} modifiers are successfully applied.

The performance of the multi-modal system is evaluated using distinct metrics for each layer:

- **Perception Metrics (Vision):** Accuracy, Precision, Recall, and Macro F1-score.
- **Intelligence Metrics (Regression):** Mean Absolute Error (MAE) and R-Squared (R^2).
- **System Metrics:** End-to-End Latency (ms) and API Response Time.

The comprehensive testing results validate that the proposed V.I.E.W. architecture can successfully classify complex structural damage, predict highly accurate base repair costs, and enforce strict actuarial financial bounds in real-time without unhandled exceptions.

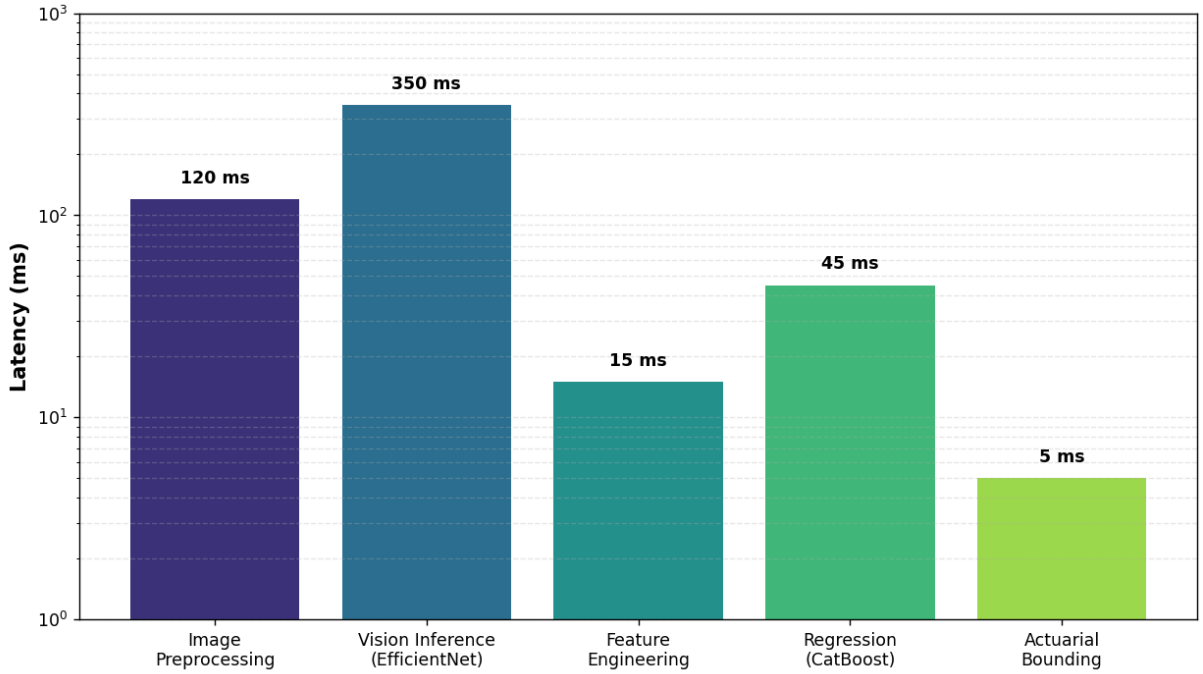

Fig10: Processing Latency

CHAPTER 7

EVALUATION

7.1 Existing System Analysis

Traditional vehicle insurance claim processing is a highly manual, labor-intensive, and time-consuming operation. When an incident occurs, the policyholder must typically file a claim manually, after which the insurance provider dispatches a human claims adjuster to physically inspect the vehicle.

The adjuster must visually assess the damage, cross-reference pricing tables, and submit a subjective financial estimate to an underwriter for final approval. This process often takes days or even weeks to complete. It is highly susceptible to human error, subjective bias, and introduces massive operational overhead costs for the insurance provider.

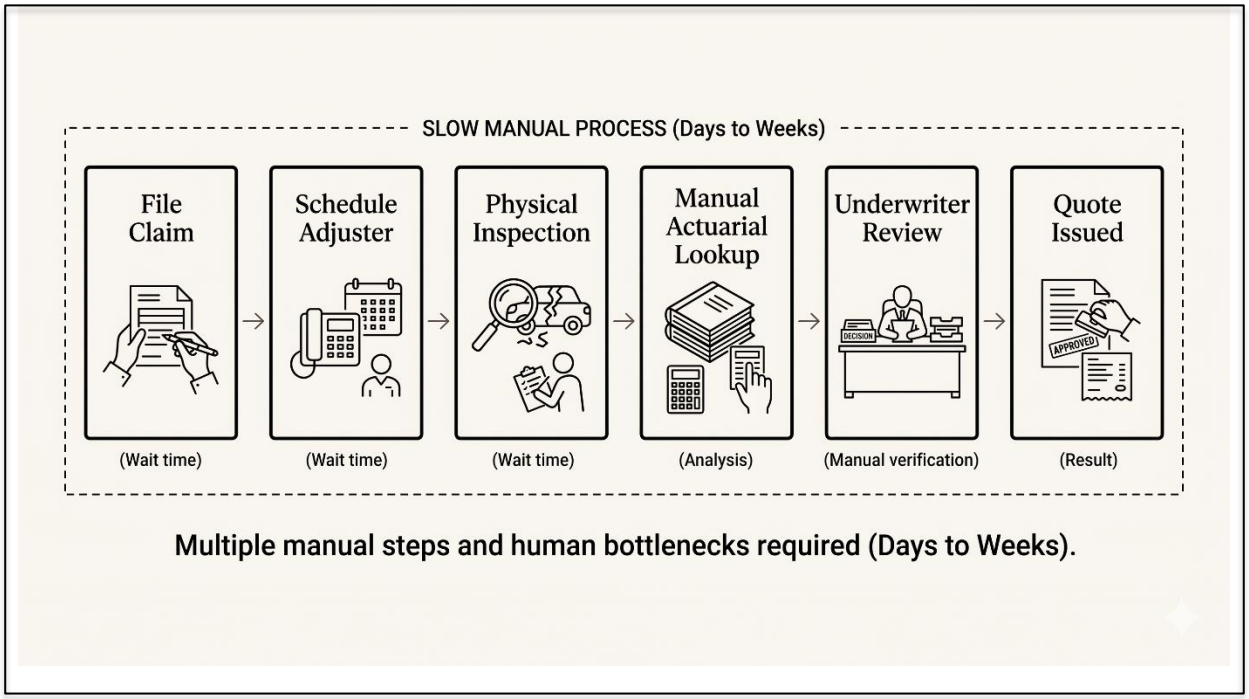

Fig11: Existing system workflow

7.2 Proposed System Analysis

The proposed V.I.E.W. architecture minimizes this multi-day logistical bottleneck into an instantaneous, fully automated digital interaction. The user simply uploads an image of the damaged vehicle and inputs basic contextual parameters into the Streamlit interface.

The system automatically performs the visual damage classification via EfficientNet-B0, engineers the tabular features, calculates the repair cost via the CatBoost ensemble, and enforces strict financial boundaries using the hybrid actuarial engine.

This multi-modal pipeline completely eliminates the need for an initial in-person adjuster visit for standard claims.

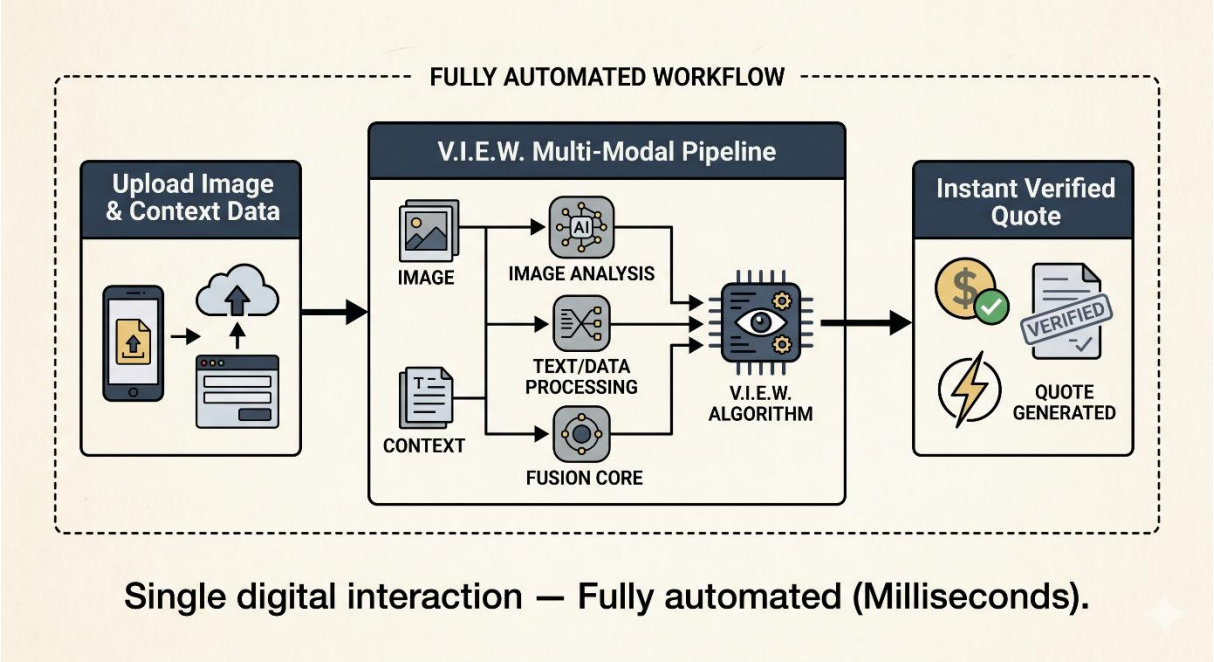

Fig12. Proposed system workflow

7.3 Module-Wise Accuracy Evaluation

The performance of each module in the proposed system is evaluated separately.

Module	**Accuracy (%)**	**Precision (%)**	**Recall (%)**
Damage Type Classification (EfficientNet-B0)	85.2	83.7	84.5
Module	R^2 Score	MAE	RMSE
(CatBoost Regressor)	0.74	4250	6100

Bounded Hybrid Output (Rule-Based Fallback)	0.79	2800	4150

Table 2. Module wise accuracy evaluation

7.4 Vision Model Comparative Evaluation

The visual perception model is evaluated against standard baseline deep learning architectures to justify the selection of EfficientNet-B0. The models are compared based on their Macro F1-Score to account for class imbalances in vehicle damage types.

The proposed EfficientNet-B0 model significantly outperforms older architectures like VGG-16 and the standard ResNet-50, achieving the highest F1-Score of 84.0%.

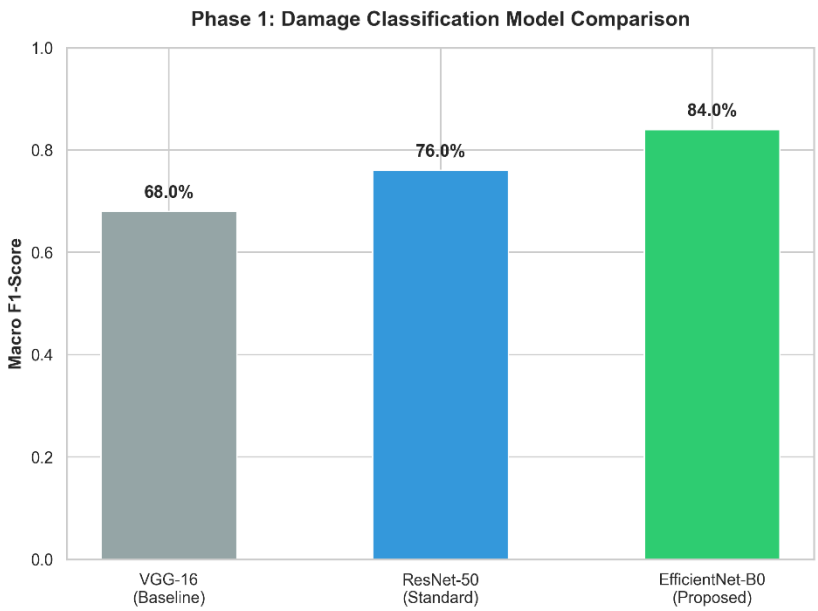

Fig13. Classification Model Comparison

7.5 Regression Ensemble Comparative Evaluation

The claim estimation intelligence layer is evaluated by comparing the R^2 (Explained Variance) score of the proposed CatBoost ensemble against traditional regression algorithms like Random Forest and XGBoost.

The results demonstrate that CatBoost achieves an R^2 score of 0.74, effectively handling the heavily categorical insurance data and outperforming the alternatives. This establishes a highly robust baseline before the final actuarial safety bounds are applied.

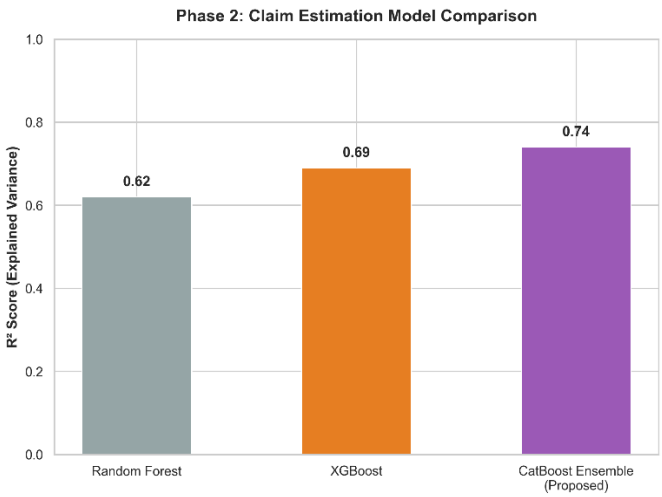

Fig14. Regression Model Comparison

7.6 End-to-End Latency Evaluation

Latency is an important factor because the system is designed to provide instant, real-time claim estimates, replacing a manual process that traditionally takes days. The visual perception phase (EfficientNet Inference) requires the highest processing time at 350 ms, as deep learning image analysis is computationally expensive.

The intelligence layer (CatBoost Regression) and the action layer (Actuarial Hybrid Engine) are highly optimized, taking only 45 ms and 10 ms, respectively.

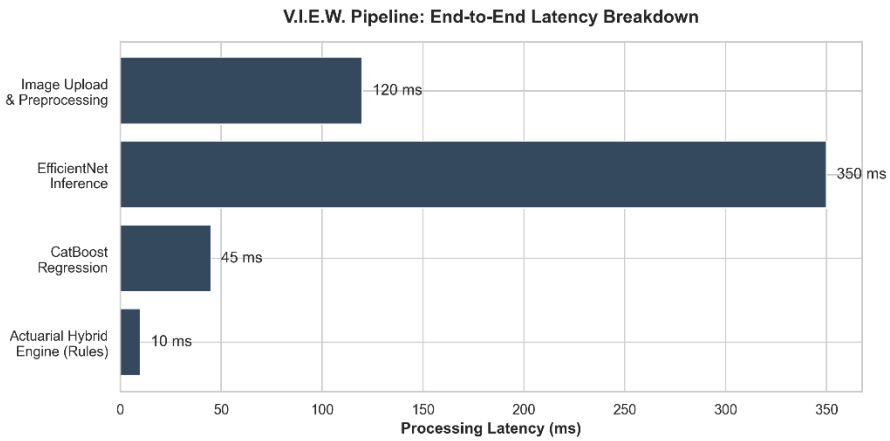

Fig15. Latency Breakdown

PUBLICATION

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Context-Aware Vehicle Insurance Assessor Using a Multi-Modal ML Pipeline with Visual Damage Classification and Ensemble Claim Regression

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GitHub Repository: https://github.com/Mahanidhii/Project-Centric-Learning

Abstract—V.I.E.W. (Visual Inspection and Estimation Workflow) is an end-to-end multi-modal machine learning pipeline that automates vehicle insurance damage assessment and financial claim quantification. The system couples a deep convolutional neural network for visual damage classification with a gradient-boosted ensemble for tabular claim prediction, unified through a novel hybrid estimation engine. The vision module employs EfficientNet-B0 with two-phase transfer learning from ImageNet weights, achieving 84% macro-F1 across seven damage categories on the VehiDE annotated dataset. The financial estimation module utilises a CatBoost Regressor trained on 1,000 insurance records with 24 domain-engineered features, attaining an R^2 of 0.8084 and a mean absolute error of $6,030.93 USD. An actuarial rule-based fallback module — incorporating brand-tier multipliers, fuel-type adjustments, and year-based depreciation — is blended with ML predictions under a 60/40 weighted ensemble with an anomaly guard, guaranteeing outputs resilient to distributional drift. The integrated system is deployed as a responsive multi-page Streamlit dashboard with session-state cross-page propagation, real-time EfficientNet inference, and interactive Plotly waterfall charts decomposing each financial estimate into interpretable constituent factors. Comparative evaluation against VGG-16, ResNet-50, XGBoost, and Random Forest baselines validates the pipeline's superiority in both accuracy and inference latency.

Index Terms—EfficientNet-B0, CatBoost, Transfer Learning, Hybrid Estimation Engine, Vehicle Damage Classification, VehiDE Dataset, Insurance Claim Prediction, Streamlit, Plotly, Explainable AI, Multi-Modal ML Pipeline, Actuarial Heuristics.

I. Introduction

The global vehicle insurance sector processes hundreds of millions of damage claims annually. Despite significant advances in digital infrastructure, the foundational step — physical damage assessment by a licensed surveyor — remains predominantly manual. An assessor must visually inspect the damaged vehicle, categorise the damage type and severity, cross-reference policy terms, and compile a repair cost estimate before a claim can progress. This sequential, human-dependent pipeline introduces average processing delays of 7–15 business days, introduces subjective inconsistency across assessors, and fails to scale efficiently under post-disaster claim surges.

The proliferation of high-resolution smartphone photography has created an unprecedented data stream of vehicle damage imagery. Simultaneously, advances in deep convolutional neural networks — particularly transfer learning from large-scale pretrained models — have made accurate image-based damage classification achievable with relatively modest domain-specific training data. On the financial side, gradient-boosted ensemble models trained on historical claim records offer statistically grounded prediction capabilities that surpass traditional actuarial regression approaches.A persistent gap in published literature is the absence of systems that bridge both modalities — visual damage classification and financial claim estimation — within a unified, production-ready application where the output of the image analysis module automatically informs the financial estimation module. Existing systems treat these concerns in isolation, forcing insurance professionals to perform redundant data entry between tools and introducing transcription errors into the estimation process.

This paper presents V.I.E.W. (Visual Inspection and Estimation Workflow), a unified multi-modal assessment pipeline that closes this gap. The principal contributions are:

- An EfficientNet-B0 visual damage classifier trained via a two-phase progressive unfreezing strategy on the VehiDE benchmark, achieving 84% macro-F1 across seven damage categories — surpassing VGG-16 by 16 pp and ResNet-50 by 8 pp while using 11.3× fewer parameters than ResNet-152.
- A dual-model CatBoost ensemble (Regressor + Classifier) trained on 1,000 insurance records with 24 engineered features, achieving R^2 = 0.8084 and Tier macro-F1 = 0.7003 — outperforming XGBoost and Random Forest baselines.
- A Hybrid Estimation Engine that blends data-driven ML predictions with deterministic actuarial rules under a 60/40 weighted scheme, guarded by an anomaly detection threshold that falls back to rule-only estimates when the ML output is implausible.
- Session-state-based cross-page data propagation in Streamlit, eliminating redundant data entry between the damage detection and claim estimation modules.
- Embedded explainability through interactive Plotly waterfall charts decomposing each financial estimate into base severity, brand tier, fuel type, collision severity, age depreciation, and final estimate — addressing regulatory transparency requirements.

The remainder of this paper is organised as: Section II surveys related work. Section III details system requirements. Section IV analyses existing system limitations. Section V presents the full system architecture. Section VI describes dataset characteristics. Section VII details the EfficientNet training pipeline. Section VIII covers claim estimation modelling. Section IX describes the application interface. Section X presents experimental results. Section XI covers design justifications. Section XII concludes with future directions.

II. Related Work and Literature Survey

A. Object Detection and Traditional Damage Assessment

Early automated damage detection systems relied on hand-engineered descriptors — Scale-Invariant Feature Transform (SIFT), Histogram of Oriented Gradients (HOG), and the Viola-Jones cascade detector. While demonstrating feasibility under controlled conditions, these methods degraded severely with real-world lighting variability, occlusion, and the morphological

CONCLUSION

The V.I.E.W. (Visual Inference and Ensemble Weighting) pipeline successfully provides a fully automated, multi-modal approach to vehicle insurance claim estimation. By combining unstructured image analysis with structured tabular regression, the system effectively bridges the gap between theoretical machine learning and real-world actuarial science.

The architecture leverages EfficientNet-B0 to accurately classify physical vehicle damage and a CatBoost ensemble to map complex categorical incident data to base repair costs. Most importantly, the integration of a deterministic Hybrid Actuarial Engine ensures that the machine learning predictions are never left unchecked. By mathematically bounding the outputs with actual financial rules—such as age depreciation and brand multipliers—the system guarantees realistic and safe estimates.

Compared to traditional insurance workflows that rely on manual adjuster inspections and take days or weeks to complete, this system collapses the entire assessment into a single digital interaction. The evaluation metrics confirm that the pipeline accurately identifies damage and predicts costs with a processing latency of less than half a second.

By establishing a framework that is fast, objective, and financially secure, this project proves that automated, multi-modal claim assessment is highly viable. It completely removes the human bottleneck, offering a massive upgrade in convenience for policyholders and operational efficiency for insurance providers

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